

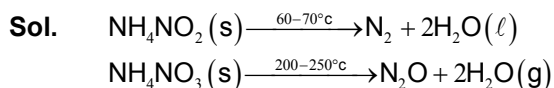
Chemistry

SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

- Q.1** The heating of NH_4NO_2 at $60\text{--}70^\circ\text{C}$ and NH_4NO_3 at $200\text{--}250^\circ\text{C}$ is associated with the formation of nitrogen containing compounds **X** and **Y**, respectively. **X** and **Y**, respectively, are
- (A) N_2 and N_2O (B) NH_3 and NO_2
 (C) NO and N_2O (D) N_2 and NH_3

Ans. A



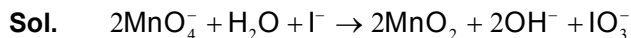
- Q.2** The correct order of the wavelength maxima of the absorption band in the ultraviolet -visible region for the given complexes is
- (A) $[\text{Co}(\text{CN})_6]^{3-} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+} < [\text{Co}(\text{NH}_3)_5(\text{Cl})]^{2+}$
 (B) $[\text{Co}(\text{NH}_3)_5(\text{Cl})]^{2+} < [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{CN})_6]^{3-}$
 (C) $[\text{Co}(\text{CN})_6]^{3-} < [\text{Co}(\text{NH}_3)_5(\text{Cl})]^{2+} < [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+}$
 (D) $[\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{CN})_6]^{3-} < [\text{Co}(\text{NH}_3)_5(\text{Cl})]^{2+} < [\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$

Ans. A

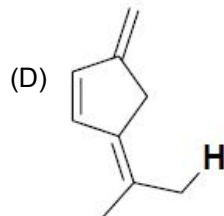
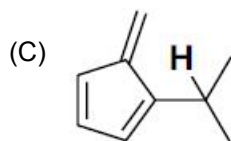
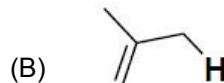
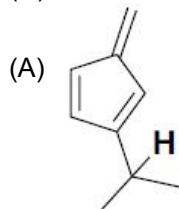
Sol. $\text{CN}^- > \text{NH}_3 > \text{H}_2\text{O} > \text{Cl}^-$, (Decreasing ligand strength order.), Strong field ligands cause larger splitting and absorb shorter wave length.

- Q.3** One of the products formed from the reaction of permanganate ion with iodide ion in neutral aqueous medium is
- (A) I_2 (B) IO_3^-
 (C) IO_4^- (D) IO_2^-

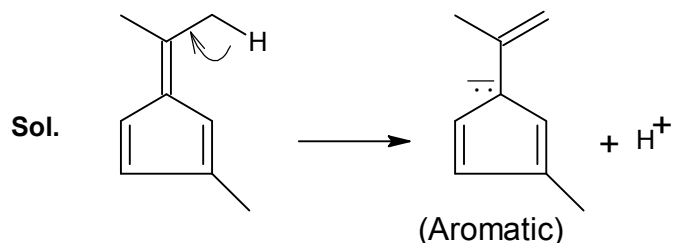
Ans. B



*Q.4 Consider the depicted hydrogen (H) in the hydrocarbons given below. The most acidic hydrogen (H) is



Ans. B



SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme:**
 - Full Marks** : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
 - Partial Marks** : +3 If all the four options are correct but **ONLY** three options are chosen;
 - Partial Marks** : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
 - Partial Marks** : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
 - Zero Marks** : 0 If none of the option is chosen (i.e. the question is unanswered);
 - Negative Marks** : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to correct answers, then
 - choosing **ONLY** (A), (B) and (D) will get +4 marks;
 - choosing **ONLY** (A) and (B) will get +2 marks;
 - choosing **ONLY** (A) and (D) will get +2marks;
 - choosing **ONLY** (B) and (D) will get +2 marks;
 - choosing **ONLY** (A) will get +1 mark;
 - choosing **ONLY** (B) will get +1 mark;
 - choosing **ONLY** (D) will get +1 mark;
 - choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
 - choosing any other option(s) will get -2 marks.

***Q.5** Regarding the molecular orbital (MO) energy levels for homonuclear diatomic molecules, the **INCORRECT** statement(s) is(are)

- (A) Bond order of Ne_2 is zero.
 (B) The highest occupied molecular orbital (HOMO) of F_2 is σ -type.
 (C) Bond energy of O_2^+ is smaller than the bond energy of O_2 .
 (D) Bond length of Li_2 is larger than the bond length of B_2 .

Ans. B, C

Sol. (B) $\rightarrow \text{F}_2 = \sigma 1s^2, \sigma 1s^{2*}, \sigma 2s^2, \sigma 2s^{2*}, \sigma 2p_x^2 \left| \begin{smallmatrix} \pi 2p_y^2 \\ \pi 2p_z^2 \end{smallmatrix} \right| \left| \begin{smallmatrix} \pi 2p_y^{2*} \\ \pi 2p_z^{2*} \end{smallmatrix} \right| \sigma 2p_x^*$

HOMO of F_2 is π -type

(C) $\rightarrow \text{O}_2 = \sigma 1s^2, \sigma 1s^{2*}, \sigma 2s^2, \sigma 2s^{2*}, \sigma 2p_x^2 \left| \begin{smallmatrix} \pi 2p_y^2 \\ \pi 2p_z^2 \end{smallmatrix} \right| \left| \begin{smallmatrix} \pi 2p_y^{1*} \\ \pi 2p_z^{1*} \end{smallmatrix} \right| \sigma 2p_x^*$

$$\text{B.O.} = \frac{1}{2} |10 - 6| = 2$$

$\text{O}_2^+ = \sigma 1s^2, \sigma 1s^{2*}, \sigma 2s^2, \sigma 2s^{2*}, \sigma 2p_x^2 \left| \begin{smallmatrix} \pi 2p_y^2 \\ \pi 2p_z^2 \end{smallmatrix} \right| \left| \begin{smallmatrix} \pi 2p_y^{1*} \\ \pi 2p_z^{1*} \end{smallmatrix} \right| \sigma 2p_x^*$

$$\text{B.O.} = \frac{1}{2} |10 - 5| = 2.5$$

Bond energy of $\text{O}_2^+ >$ Bond energy of O_2

$\rightarrow \text{Li}_2$ and B_2 has same bond order, but B_2 has smaller atomic size so it has smaller bond length

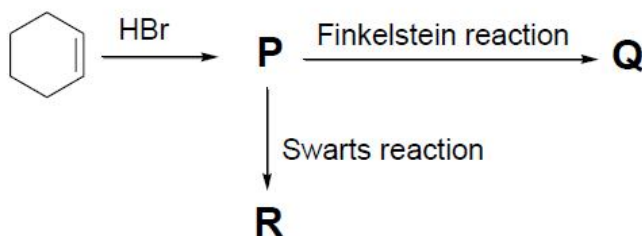
Q.6 The pair(s) of diamagnetic ions is(are)

- (A) $\text{La}^{3+}, \text{Ce}^{4+}$ (B) $\text{Yb}^{2+}, \text{Lu}^{3+}$
 (C) $\text{La}^{2+}, \text{Ce}^{3+}$ (D) $\text{Yb}^{3+}, \text{Lu}^{2+}$

Ans. A, B

Sol. $\text{La} = [\text{Xe}] 5d^1, 6s^2$, $\text{La}^{3+} = [\text{Xe}]$ (Diamagnetic)
 $\text{Ce} = [\text{Xe}] 4f^1 5d^1, 6s^2$, $\text{Ce}^{4+} = [\text{Xe}]$ (Diamagnetic)
 $\text{Yb} = [\text{Xe}] 4f^{14}, 6s^2$, $\text{Yb}^{2+} = [\text{Xe}], 4f^{14}$ (Diamagnetic)
 $\text{Lu} = [\text{Xe}] 4f^{14}, 5d^1, 6s^2$, $\text{Lu}^{3+} = [\text{Xe}] 4f^{14}$ (Diamagnetic)

Q.7 For the reaction sequence given below, the correct statement(s) is(are)

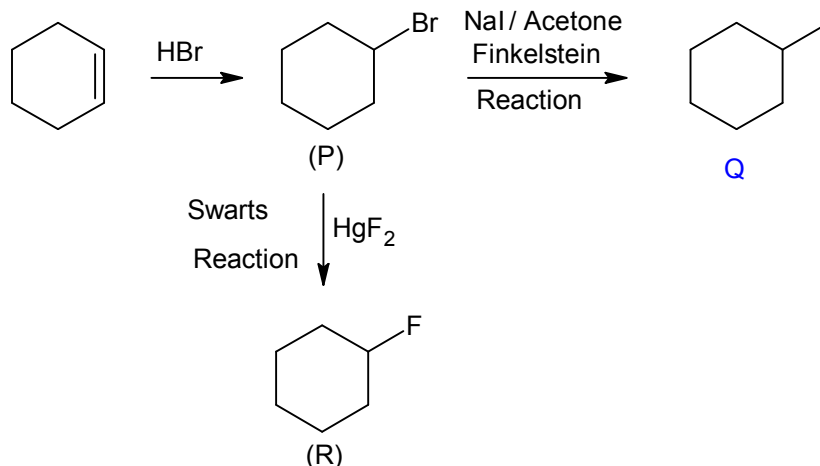


(In the options, X is any atom other than carbon and hydrogen, and it is different in **P**, **Q** and **R**)

- (A) C-X bond length in **P**, **Q** and **R** follows the order **Q** > **R** > **P**.
 (B) C-X bond enthalpy in **P**, **Q** and **R** follows the order **R** > **P** > **Q**.
 (C) Relative reactivity toward $\text{S}_{\text{N}}2$ reaction in **P**, **Q** and **R** follows the order **P** > **R** > **Q**.
 (D) pK_{a} value of the conjugate acids of the leaving groups in **P**, **Q** and **R** follows the order **R** > **Q** > **P**.

Ans. B

Sol.



C – X bond enthalpy in P, Q, R follows the order are $R > P > Q$

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks : +4 If **ONLY** the correct integer is entered in the designated place;
Zero Marks : 0 In all other cases.

Q. 8 In an electrochemical cell, dichromate ions in aqueous acidic medium are reduced to Cr^{3+} . The current (in amperes) that flows through the cell for 48.25 minutes to produce 1 mole of Cr^{3+} is _____.

Use: 1 Faraday = 96500 C mol^{-1}

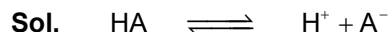
Ans. 100.00

Sol. $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$
 2 mole Cr^{3+} is produced by = 6 mole e^-
 1 mole Cr^{3+} is produced by = 3 mole e^-
 Charge = $i \times t$
 $3 \times 96500 = i \times 48.25 \times 60$
 $i = 100 \text{ amp}$

***Q. 9** At 25°C , the concentration of H^+ ions in $1.00 \times 10^{-3} \text{ M}$ aqueous solution of a weak monobasic acid having acid dissociation constant (K_a) of 4.00×10^{-11} is $X \times 10^{-7} \text{ M}$. The value of **X** is _____.

Use: Ionic product of water (K_w) = 1.00×10^{-14} at 25°C .

Ans. 2.24



$$\begin{array}{ccc} 10^{-3} & 0 & 0 \\ (10^{-3} - x) & (x+y) & x \end{array}$$



$$\begin{array}{ccc} & (y+x) & y \end{array}$$

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = \frac{(x+y) \cdot x}{(10^{-3} - x)}$$

$$4 \times 10^{-11} = \frac{(x+y)x}{10^{-3}}, [x \ll 10^{-3}]$$

$$\text{or } x(x+y) = 4 \times 10^{-14} \quad (\text{I})$$

$$\text{also } y(x+y) = 1 \times 10^{-14} \quad (\text{II})$$

$$\text{On solving } x = 1.788 \times 10^{-7}$$

$$y = 0.447 \times 10^{-7}$$

$$\therefore x + y = 2.235 \times 10^{-7}$$

$$[\text{H}^+]_{\text{T}} = 2.24 \times 10^{-7}$$

$$x = 2.24$$

- *Q. 10** Molar volume (V_m) of a van der Waals gas can be calculated by expressing the van der Waals equation as a cubic equation with V_m as the variable. The ratio (in mol dm^{-3}) of the coefficient of V_m^2 to the coefficient of V_m for a gas having van der Waals constant $a = 6.0 \text{ dm}^6 \text{ atm mol}^{-2}$ and $b = 0.060 \text{ dm}^3 \text{ mol}^{-1}$ at 300 K and 300 atm is _____.
Use: Universal gas constant (R) = $0.082 \text{ dm}^3 \text{ atm mol}^{-1} \text{ K}^{-1}$.

Ans. -7.1

Sol. $\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$, (For 1 mole)

$$PV_m - Pb + \frac{a}{V_m} - \frac{ab}{V_m^2} - RT = 0$$

$$PV_m^3 - V_m^2 Pb + aV_m - ab - RTV_m^2 = 0$$

$$PV_m^3 - V_m^2 (Pb + RT) + V_m a - ab = 0$$

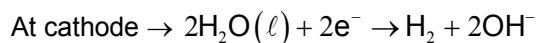
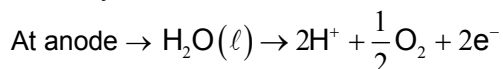
$$\text{Ratio} = \frac{-(Pb + RT)}{a} = \frac{(300 \times 0.06 + 0.082 \times 300)}{6}$$

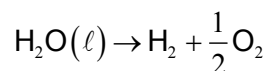
$$= -7.1$$

- *Q. 11** Considering ideal gas behavior, the expansion work done (in kJ) when 144 g of water is electrolyzed completely under constant pressure at 300 K is _____.
Use: Universal gas constant (R) = $8.3 \text{ J K}^{-1} \text{ mol}^{-1}$; Atomic mass (in amu) : H = 1, O = 16

Ans. 29.88

Sol. Electrolysis of H_2O





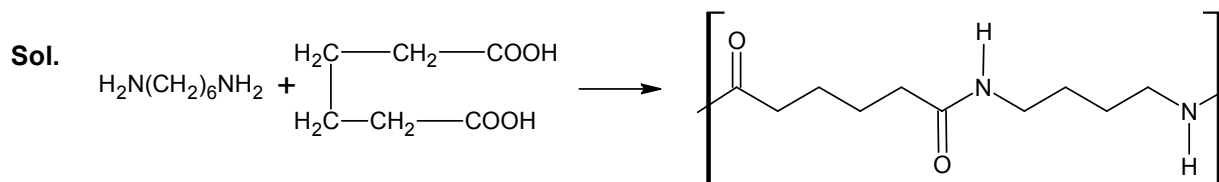
8 mole H_2O will provide = 8 mole H_2 & 4 mole O_2 , total moles of gases produced = 12 mole

$$\begin{aligned}\text{Work done} &= -P_{\text{ext}} \times \Delta V \\ &= -\Delta n_g RT \\ &= -12 \times 8.3 \times 300 \\ &= -29880 \text{ J} \\ &\text{Or } 29.880 \text{ KJ}\end{aligned}$$

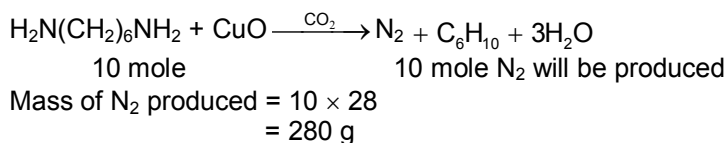
- Q. 12** The monomer (**X**) involved in the synthesis of Nylon 6,6 gives positive carbylamine test. If 10 moles of **X** are analyzed using Dumas method, the amount (in grams) of nitrogen gas evolved is _____.

Use: Atomic mass of N (in amu) = 14

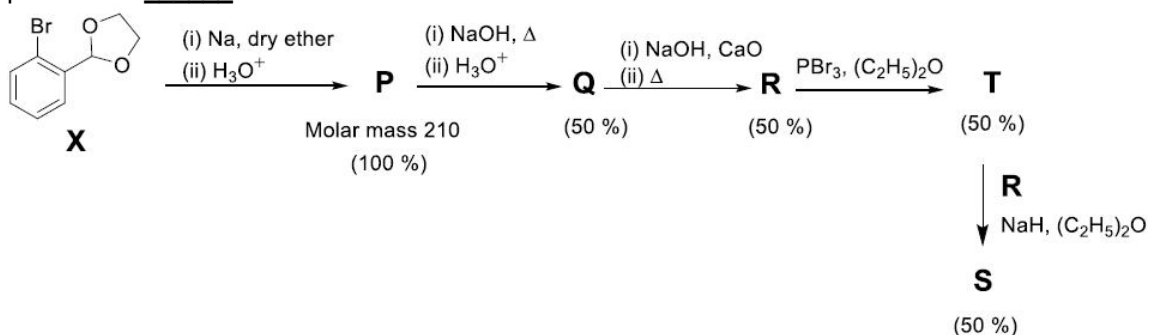
Ans. 280



Nylon 66



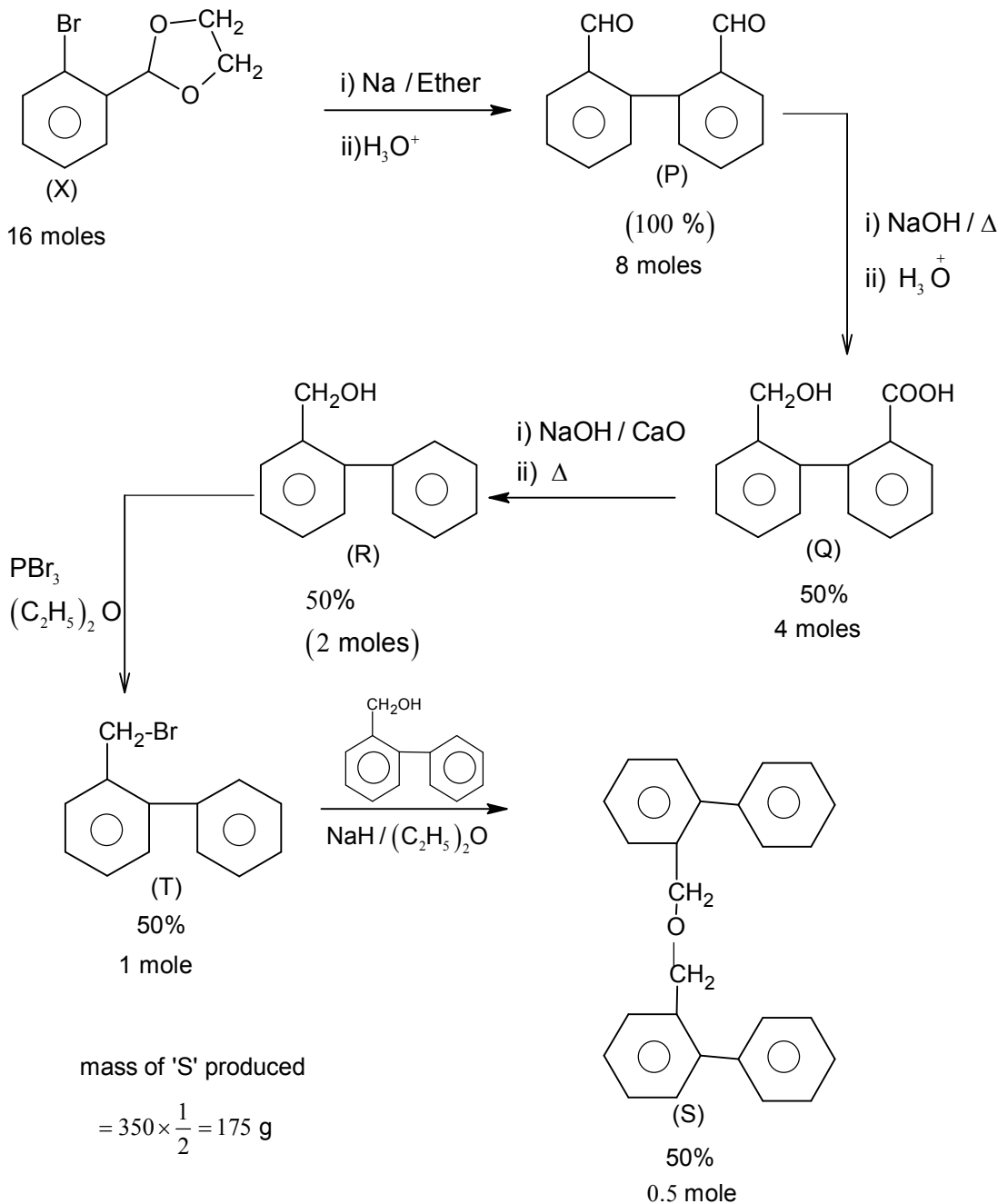
- Q. 13** The reaction sequence given below is carried out with 16 moles of **X**. The yield of the major product in each step is given below the product in parentheses. The amount (in grams) of **S** produced is _____.



Use: Atomic mass (in amu) : H = 1, C = 12, O = 16, Br = 80

Ans. 175

Sol.

**SECTION 4 (Maximum Marks: 12)**

- This section contains **THREE (03)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks	: +4	ONLY if the option corresponding to the correct combination is chosen;
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	: -1	In all other cases.

- Q.14** The correct match of the group reagents in **List-I** for precipitating the metal ion given in **List-II** from solutions, is

List-I		List-II	
(P)	Passing H_2S in the presence of NH_4OH	(1)	Cu^{2+}
(Q)	$(\text{NH}_4)_2\text{CO}_3$ in the presence of NH_4OH	(2)	Al^{3+}
(R)	NH_4OH in the presence of NH_4Cl	(3)	Mn^{2+}
(S)	Passing H_2S in the presence of dilute HCl	(4)	Ba^{2+}
		(5)	Mg^{2+}

- (A) $\text{P} \rightarrow 3; \text{Q} \rightarrow 4; \text{R} \rightarrow 2; \text{S} \rightarrow 1$
 (B) $\text{P} \rightarrow 4; \text{Q} \rightarrow 2; \text{R} \rightarrow 3; \text{S} \rightarrow 1$
 (C) $\text{P} \rightarrow 3; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 5$
 (D) $\text{P} \rightarrow 5; \text{Q} \rightarrow 3; \text{R} \rightarrow 2; \text{S} \rightarrow 4$

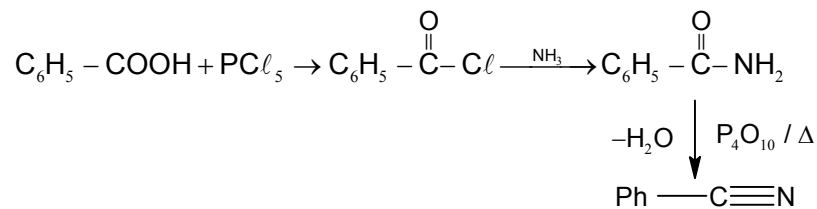
Ans. A

Sol. $\text{P} - 3$
 H_2S in the presence of NH_4OH precipitates Mn^{2+} i.e. group IV cation
 $\text{Q} - 4$
 $(\text{NH}_4)_2\text{CO}_3$ in the presence of NH_4OH precipitate Ba^{++} i.e. group V cation
 $\text{R} - 2$
 NH_4OH in the presence of NH_4Cl precipitate Al^{3+} i.e. group III cation
 $\text{S} - 1$
 H_2S in presence of HCl will ppt. Cu^{++} , i.e. group II cation.

- Q.15** The major products obtained from the reactions in **List-II** are the reactants for the named reactions mentioned in **List-I**. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

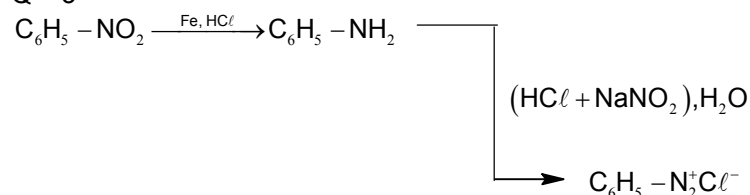
List-I		List-II	
(P)	Stephen reaction	(1)	Toluene $\xrightarrow[\text{(ii) } \text{H}_3\text{O}^+]{\text{(i) } \text{CrO}_2\text{Cl}_2 / \text{CS}_2}$
(Q)	Sandmeyer reaction	(2)	Benzoic acid $\xrightarrow[\text{(iii) } \text{P}_4\text{O}_{10}, \Delta]{\text{(i) } \text{PCl}_5, \text{(ii) } \text{NH}_3}$
(R)	Hoffmann bromamide degradation reaction	(3)	Nitrobenzene $\xrightarrow[\text{(273-278 K), } \text{H}_2\text{O}]{\text{(i) } \text{Fe, HCl}, \text{(ii) } \text{HCl, NaNO}_2}$
(S)	Cannizzaro reaction	(4)	Toluene $\xrightarrow[\text{(iv) } \text{NH}_3]{\text{(i) } \text{Cl}_2 / \text{h}\nu, \text{H}_2\text{O}, \text{(ii) Tollen's reagent}, \text{(iii) } \text{SO}_2\text{Cl}_2}$
		(5)	Aniline $\xrightarrow[\text{(iii) aq. NaOH}]{\text{(i) } (\text{CH}_3\text{CO})_2\text{O, Pyridine}, \text{(ii) } \text{HNO}_3, \text{H}_2\text{SO}_4, 288 \text{ K}}$

- (A) $\text{P} \rightarrow 2; \text{Q} \rightarrow 4; \text{R} \rightarrow 1; \text{S} \rightarrow 3$
 (B) $\text{P} \rightarrow 2; \text{Q} \rightarrow 3; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

(C) $P \rightarrow 5; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$ (D) $P \rightarrow 5; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$ **Ans. B****Sol.** P – 2

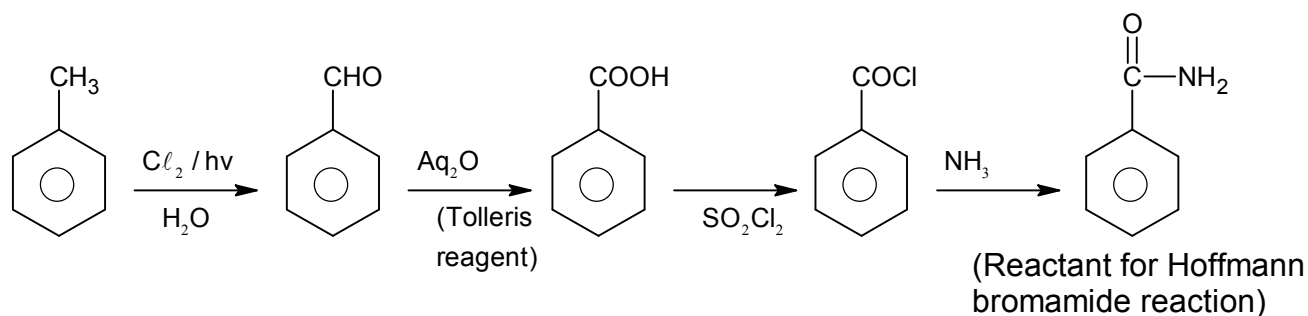
(Reactant for Stephen reaction)

Q – 3

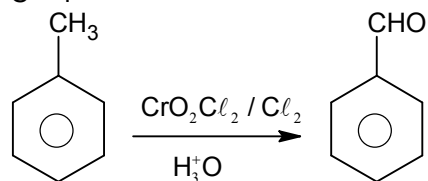


(Reactant for Sandmeyer reaction)

R – 4

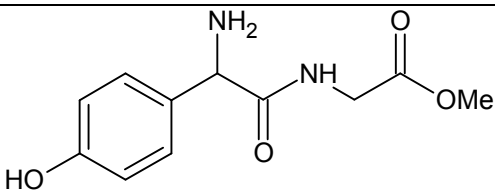
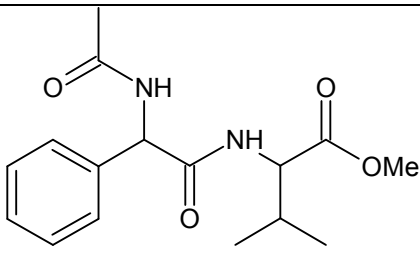
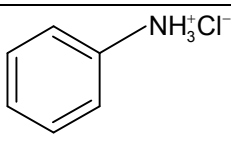
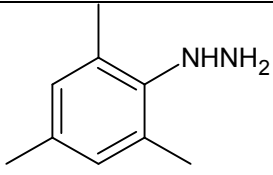


S – 1



(Reactant for Cannizzaro reaction)

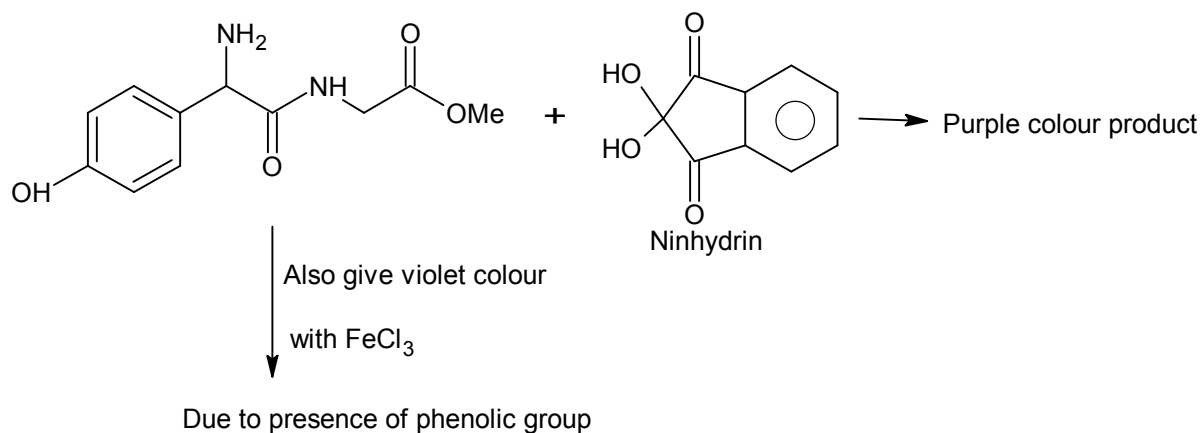
Q.16 Match the compounds in **List-I** with the appropriate observations in **List-II** and choose the correct option.

List-I	List-II
(P) 	(1) Reaction with phenyl diazonium salt gives yellow dye.
(Q) 	(2) Reaction with ninhydrin gives purple color and it also reacts with FeCl_3 to give violet color.
(R) 	(3) Reaction with glucose will give corresponding hydrazone.
(S) 	(4) Lassaigne extract of the compound treated with dilute HCl followed by addition of aqueous FeCl_3 gives blood red color.
	(5) After complete hydrolysis, it will give ninhydrin test and it DOES NOT give positive phthalein dye test.

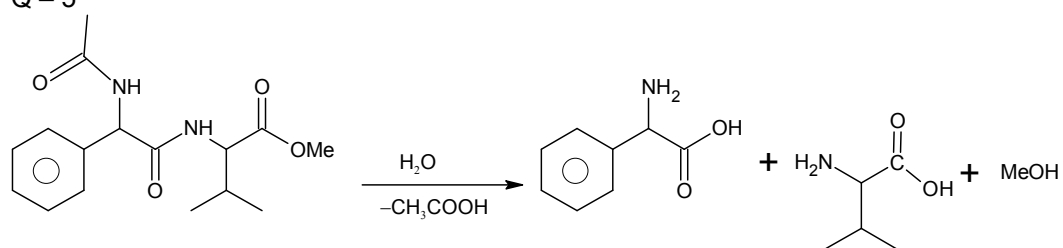
- (A) $\text{P} \rightarrow 1; \text{Q} \rightarrow 5; \text{R} \rightarrow 4; \text{S} \rightarrow 2$
 (B) $\text{P} \rightarrow 2; \text{Q} \rightarrow 5; \text{R} \rightarrow 1; \text{S} \rightarrow 3$
 (C) $\text{P} \rightarrow 5; \text{Q} \rightarrow 2; \text{R} \rightarrow 1; \text{S} \rightarrow 4$
 (D) $\text{P} \rightarrow 2; \text{Q} \rightarrow 1; \text{R} \rightarrow 5; \text{S} \rightarrow 3$

Ans. B

Sol. P – 2



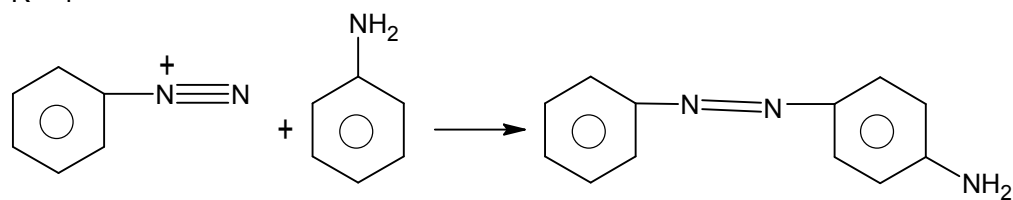
Q – 5



(Gives Ninhydrin test)

Does not give phthalein test due to absence of phenolic group)

R – 1



Yellow dye

S – 3

